CLASS 12 MATHEMATICS NOTES & PRACTICE

Topic: Bayes' Theorem & Random Variables / Probability Distributions

1. PRACTICE PROBLEMS: BAYES' THEOREM

Bayes' Theorem finds the conditional probability of an event based on prior knowledge of conditions related to the event. Formula:

$$P(E_i|A) = [\, P(E_i) \cdot P(A|E_i) \,] / [\, \Sigma_{k=1}^{n} P(E_k) \cdot P(A|E_k) \,]$$

Problem 1 (Bag & Ball Selection)

Bag I contains 3 red and 4 black balls, while Bag II contains 5 red and 6 black balls. One ball is drawn at random from one of the bags and it is found to be red. Find the probability that it was drawn from Bag II.

Solution:

Let E_1 = Event of choosing Bag I, E_2 = Event of choosing Bag II, and A = Event of drawing a red ball.

$P(E_1) = 1/2$, $P(E_2) = 1/2$

$P(A|E_1) = 3/7$ (Red from Bag I), $P(A|E_2) = 5/11$ (Red from Bag II)

By Bayes' Theorem:

$$P(E_2|A) = [\, (1/2) \times (5/11) \,] / [\, (1/2 \times 3/7) + (1/2 \times 5/11) \,] = (5/11) / (3/7 + 5/11) = (5/11) / (68/77) = 35/68$$

Problem 2 (Factory Machine Production)

In a factory, Machine A produces 60% of items and Machine B produces 40%. 2% of items from Machine A and 1% from Machine B are defective. An item is chosen at random and found defective. Find the probability it was produced by Machine A.

Solution:

Let E_1 = Item produced by Machine A, E_2 = Item produced by Machine B, and D = Item is defective.

$P(E_1) = 0.60$, $P(E_2) = 0.40$

$P(D|E_1) = 0.02$, $P(D|E_2) = 0.01$

By Bayes' Theorem:

$$P(E_1|D) = (0.60 \times 0.02) / [\, (0.60 \times 0.02) + (0.40 \times 0.01) \,] = 0.012 / (0.012 + 0.004) = 0.012 / 0.016 = 3/4$$

Problem 3 (Coin Selection: Double-Headed vs Normal)

Given three coins: one is a two-headed coin, another is a biased coin that comes up heads 75% of the time, and the third is an unbiased coin. One coin is chosen at random and tossed; it shows heads. What is the probability that it was the two-headed coin?

Solution:

Let E_1 = Two-headed coin, E_2 = Biased coin, E_3 = Unbiased coin, and H = Coin shows heads.

$P(E_1) = P(E_2) = P(E_3) = 1/3$

$P(H|E_1) = 1$, $P(H|E_2) = 3/4$, $P(H|E_3) = 1/2$

By Bayes' Theorem:

$$P(E_1|H) = (1/3 \times 1) / [(1/3 \times 1) + (1/3 \times 3/4) + (1/3 \times 1/2)] = 1 / (1 + 3/4 + 1/2) = 1 / (9/4) = 4/9$$

Problem 4 (Multiple Choice Test)

In answering a question on a multiple-choice test, a student either knows the answer or guesses. Let $3/4$ be the probability that he knows the answer and $1/4$ be the probability that he guesses. Assuming that a student who guesses gets the answer right with probability $1/4$. What is the probability that the student knows the answer given that he answered it correctly?

Solution:

Let E_1 = Student knows the answer, E_2 = Student guesses, and A = Student answers correctly.

$P(E_1) = 3/4$, $P(E_2) = 1/4$

$P(A|E_1) = 1$ (If he knows, he answers correctly), $P(A|E_2) = 1/4$

By Bayes' Theorem:

$$P(E_1|A) = (3/4 \times 1) / [(3/4 \times 1) + (1/4 \times 1/4)] = (3/4) / (3/4 + 1/16) = (3/4) / (13/16) = 12/13$$

Problem 5 (Medical Diagnosis Test)

A doctor knows that 1% of a population has a specific disease. A diagnostic test correctly identifies the disease 99% of the time for sick individuals, but yields a false positive 2% of the time for healthy individuals. If a randomly chosen person tests positive, what is the probability they actually have the disease?

Solution:

Let E_1 = Person has disease, E_2 = Person is healthy, and T = Test is positive.

$P(E_1) = 0.01$, $P(E_2) = 0.99$

$P(T|E_1) = 0.99$, $P(T|E_2) = 0.02$

By Bayes' Theorem:

$$P(E_1|T) = (0.01 \times 0.99) / [(0.01 \times 0.99) + (0.99 \times 0.02)] = 0.0099 / (0.0099 + 0.0198) = 0.0099 / 0.0297 = 1/3$$

2. REVISION NOTES: RANDOM VARIABLES & PROBABILITY DISTRIBUTIONS

A. Random Variable (X)

A **Random Variable** is a real-valued function whose domain is the sample space S of a random experiment. In simple terms, it assigns a numerical value to each outcome of an experiment.

- **Discrete Random Variable:** Takes isolated finite or countable infinite values (e.g., number of heads in 3 coin tosses).
- **Continuous Random Variable:** Takes any value within a given interval (e.g., height, weight, time).

B. Probability Distribution of a Discrete Random Variable

The system of values of a random variable X together with their corresponding probabilities $P(X = x_i) = p_i$ is called the probability distribution of X.

Essential Properties:

1. $p_i \geq 0$ for all i
2. $\Sigma p_i = 1$ (Sum of all probabilities must equal 1)

C. Mean and Variance of a Probability Distribution

- **Mean (Expectation, $E(X)$ or μ):**

$$E(X) = \mu = \Sigma (x_i \cdot p_i)$$

- **Variance ($Var(X)$ or σ^2):**

$$Var(X) = E(X^2) - [E(X)]^2 = \Sigma (x_i^2 \cdot p_i) - (\mu)^2$$

- **Standard Deviation (σ):**

$$\sigma = \sqrt{[Var(X)]}$$

3. SOLVED EXAMPLES ON RANDOM VARIABLES

Example 1: Finding Probability Distribution & Expectation

Two cards are drawn successively with replacement from a well-shuffled deck of 52 cards. Find the probability distribution and mean of the number of aces.

Solution:
Let X = Number of aces drawn in 2 draws. X can take values $0, 1, 2$.
Probability of getting an ace, $p = 4/52 = 1/13$.

Probability of not getting an ace, $q = 1 - 1/13 = 12/13$.

- $P(X = 0) = q \times q = (12/13) \times (12/13) = 144/169$
- $P(X = 1) = 2 \times p \times q = 2 \times (1/13) \times (12/13) = 24/169$
- $P(X = 2) = p \times p = (1/13) \times (1/13) = 1/169$

Probability Table:

$X\ (x_i)$	0	1	2
$P(X = x_i)$	144/169	24/169	1/169

Mean $E(X)$:

$$E(X) = \Sigma\, x_i p_i = (0 \times 144/169) + (1 \times 24/169) + (2 \times 1/169) = 26/169 = 2/13$$

Example 2: Finding Constant 'k' and Variance

A random variable X has the following probability distribution:

X	0	1	2	3
$P(X)$	k	2k	3k	k^2

Find: (i) Value of k (ii) $P(X \geq 2)$

Solution:

(i) Since $\Sigma\, P(X) = 1$:

$$k + 2k + 3k + k^2 = 1 \Rightarrow k^2 + 6k - 1 = 0$$

Solving for k using quadratic formula:

$$k = [\, -6 \pm \sqrt{(36 + 4)}\,] / 2 = [\, -6 \pm \sqrt{40}\,] / 2 = -3 + \sqrt{10} \quad \text{(taking positive value since } k > 0\text{)}$$

(ii) $P(X \geq 2) = P(X = 2) + P(X = 3) = 3k + k^2$.